

Mailchimp Integration Summary

Overview

This document summarizes the Mailchimp integration that was added to the Expat Mindset Test application. The integration allows automatic subscription of users to a Mailchimp mailing list when they start the test.

Integration Date: October 27, 2025

Files Created

1. `lib/mailchimp-service.ts`

Purpose: Mailchimp API service for managing email subscriptions

Key Features:

- Subscribe users to Mailchimp mailing list
- Unsubscribe users from mailing list
- Check subscription status
- Proper error handling and logging
- Uses Mailchimp Marketing API v3.0

Functions:

- `subscribeToMailchimp(data)` - Subscribe a user with email, name, tags, and merge fields
- `unsubscribeFromMailchimp(email)` - Unsubscribe a user by email
- `checkSubscriptionStatus(email)` - Check if an email is subscribed

Environment Variables Required:

- `MAILCHIMP_API_KEY`
 - `MAILCHIMP_SERVER_PREFIX`
 - `MAILCHIMP_LIST_ID`
-

2. `app/api/mailchimp/subscribe/route.ts`

Purpose: Next.js API route for Mailchimp subscription

Endpoints:

- `POST /api/mailchimp/subscribe` - Subscribe a user to the mailing list
- `GET /api/mailchimp/subscribe` - Health check and configuration status

Request Body (POST):

```
{
  "email": "user@example.com",
  "firstName": "John",
  "tags": ["Expat Mindset Test"],
  "mergeFields": {
    "SOURCE": "Expat Mindset Test"
  }
}
```

Features:

- Input validation (email format, required fields)
 - Graceful handling when Mailchimp is not configured
 - Non-blocking errors (app continues even if Mailchimp fails)
 - Comprehensive error logging
-

Files Modified

1. components/user-form.tsx

Changes: Modified the `handleSubmit` function ONLY

What was added:

- Mailchimp subscription API call after local user session is saved
- Non-blocking try-catch to ensure quiz continues even if Mailchimp fails
- Console logging for subscription success/failure
- Tags and merge fields sent with subscription

What was NOT changed:

- ☒ All UI text and labels remain unchanged
- ☒ All form fields remain unchanged
- ☒ All validation logic remains unchanged
- ☒ All existing toast messages remain unchanged
- ☒ Navigation flow remains unchanged

Modified section (lines 26-94):

```
const handleSubmit = async (e: React.FormEvent) => {
  // ... existing code ...

  // NEW: Subscribe to Mailchimp (non-blocking)
  try {
    const mailchimpResponse = await fetch('/api/mailchimp/subscribe', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        email: email.trim().toLowerCase(),
        firstName: name.trim(),
        tags: ['Expat Mindset Test'],
        mergeFields: { SOURCE: 'Expat Mindset Test' }
      }),
    });
    // ... logging ...
  } catch (mailchimpError) {
    // Non-critical error - app continues
  }

  // ... rest of existing code ...
};
```

2. .env.example

Changes: Added Mailchimp environment variables

Added configuration:

```
# Mailchimp Configuration
# Sign up at mailchimp.com and create a mailing list
# API Key: Account > Extras > API Keys
# Server Prefix: Found in your API key (e.g., us1, us2, etc.)
# List ID: Audience > Settings > Audience name and defaults > Audience ID
MAILCHIMP_API_KEY=your_mailchimp_api_key_here
MAILCHIMP_SERVER_PREFIX=us1
MAILCHIMP_LIST_ID=your_list_id_here
```

What was NOT changed:

- ☒ Existing EmailJS configuration remains unchanged
- ☒ All existing comments and structure preserved

Integration Features

☒ Non-Blocking Design

- Mailchimp subscription is optional and non-blocking
- Quiz continues even if Mailchimp API fails
- Errors are logged but don't affect user experience

☒ Graceful Fallback

- App works without Mailchimp configuration

- If environment variables are missing, subscription is skipped
- Local user session is always saved regardless of Mailchimp status

✓ Security

- API keys stored in environment variables (server-side only)
- Email addresses normalized to lowercase
- Input validation on both client and server

✓ User Privacy

- Users are automatically subscribed when starting the test
- Subscription happens in the background
- Tag added: "Expat Mindset Test" for easy segmentation

Setup Instructions

1. Create Mailchimp Account

1. Sign up at mailchimp.com (<https://mailchimp.com>)
2. Create a new audience/mailling list

2. Get API Credentials

1. **API Key:** Account > Extras > API Keys
2. **Server Prefix:** Found in your API key (e.g., `abc123def456-us1`, the `us1` part)
3. **List ID:** Audience > Settings > Audience name and defaults > Audience ID

3. Configure Environment Variables

Create a `.env.local` file in the project root:

```
MAILCHIMP_API_KEY=your_actual_api_key_here
MAILCHIMP_SERVER_PREFIX=us1
MAILCHIMP_LIST_ID=your_actual_list_id_here
```

4. Test the Integration

1. Start the development server: `npm run dev`
2. Fill out the user form and submit
3. Check browser console for Mailchimp subscription logs
4. Verify in Mailchimp dashboard that the user was added

Testing

Test Mailchimp Configuration

```
curl http://localhost:3000/api/mailchimp/subscribe
```

Expected response:

```
{  
  "success": true,  
  "message": "Mailchimp subscribe API is running",  
  "configured": true  
}
```

Test Subscription

```
curl -X POST http://localhost:3000/api/mailchimp/subscribe \  
-H "Content-Type: application/json" \  
-d '{  
  "email": "test@example.com",  
  "firstName": "Test User"  
}'
```

Troubleshooting

Issue: “Missing required Mailchimp configuration”

Solution: Ensure all three environment variables are set in `.env.local`

Issue: “Invalid API key”

Solution:

- Verify API key is correct in Mailchimp dashboard
- Check that server prefix matches your API key

Issue: “Resource not found”

Solution: Verify the List ID is correct in Mailchimp dashboard

Issue: Subscription not appearing in Mailchimp

Solution:

- Check browser console for error messages
- Verify email format is valid
- Check Mailchimp dashboard > Audience > Recent activity

Future Enhancements

Possible improvements for future versions:

1. **Double Opt-in:** Set subscription status to ‘pending’ to require email confirmation
 2. **Custom Fields:** Add more merge fields (e.g., quiz results, date taken)
 3. **Segmentation:** Automatically segment users based on quiz results
 4. **Unsubscribe Link:** Add option to unsubscribe from mailing list
 5. **Batch Processing:** Queue subscriptions for better performance
 6. **Analytics:** Track subscription success rate and failures
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File Structure After Integration

```

Expat_mindset_test/
├── app/
│   ├── api/
│   │   ├── mailchimp/
│   │   │   ├── subscribe/
│   │   │   │   └── route.ts      NEW API ENDPOINT
│   │   ├── quiz/
│   │   └── users/
│   ├── quiz/
│   ├── results/
│   ├── globals.css
│   ├── layout.tsx
│   └── page.tsx
├── components/
│   ├── ui/
│   └── user-form.tsx            MODIFIED (handleSubmit only)
├── lib/
│   ├── emailjs-config.ts
│   ├── emailjs-service.ts
│   ├── local-storage.ts
│   ├── mailchimp-service.ts    NEW SERVICE
│   ├── questions.ts
│   ├── types.ts
│   └── utils.ts
├── .env.example                MODIFIED (added Mailchimp vars)
├── MAILCHIMP_INTEGRATION_SUMMARY.md  NEW DOCUMENTATION
└── ... (other files)

```

Summary

✓ 2 New Files Created:

- lib/mailchimp-service.ts
- app/api/mailchimp/subscribe/route.ts

✓ 2 Files Modified:

- components/user-form.tsx (handleSubmit function only)
- .env.example (added Mailchimp configuration)

✓ UI Preserved:

- All text, labels, and translations remain unchanged
- Only backend integration logic was modified

✓ Non-Breaking Integration:

- App works with or without Mailchimp configuration
- Existing functionality remains intact
- User experience unchanged

Integration completed successfully! 🎉

For questions or issues, please refer to the Mailchimp API documentation:
<https://mailchimp.com/developer/marketing/api/>